

Why pilots are worried about plans to replace co-pilots with AI

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Comment and Technology

A cost-cutting initiative in the world of passenger aviation could see flight-deck staff reduced to just a captain, with their co-pilot replaced by AI. It may save money, but it's a risk too far, argues **Paul Marks**

A dangerous idea is stalking the world of passenger aviation: that of halving, sometime in the 2030s, the number of pilots at the helm of civilian airliners and filling the vacant seats with AI – a move experts say could make flying far less safe. Instead of a captain and co-pilot on the flight deck, as big jets have today, Single Pilot Operations (SPOs) will have just one pilot alongside an AI somehow designed to undertake the tough, safety-critical role of co-piloting.

This, airlines argue, will address a pilot shortage that has become economically debilitating for the industry. But SPO is meeting stiff resistance from pilot trade unions globally, who slate the idea as a cost-cutting measure. Worse, as a conference on SPO at the UK's Royal Aeronautical Society (RAeS) heard last month, rather than improving safety, AI co-pilots might quash a host of important human-human interactions.

Specifically, say SPO's critics, an AI co-pilot will eliminate the ways the flight crew monitor each other's actions at the controls, and it won't replicate how pilots can calm each other in stressful situations – such as coping with a “startle” response after a technical or weather issue develops. Pilots can also cover for each other after a rest break, when sleep inertia means it takes half an hour to get back full alertness. Will AI need heightened capability at that time? No one knows.

To be fair, SPO's backers plan a two-stage build-up to automation: first, a variant called extended Minimum Crew Operations (eMCO) will operate, in which there are two human pilots on board, but there will be extended periods where only one is at the controls, supported by AI, while the other pilot rests. Once that is perfected, SPO proper could begin single pilot gate-to-gate flights.

Tanja Harter, president of the European Cockpit Association, a pilots' union, told the RAeS summit that SPO doesn't address a key crew threat: pilot fatigue. While one pilot is resting under eMCO, she said, “the other one, sitting alone up front, will suffer increased fatigue”. Other delegates agreed, citing boredom as a stressor. “Two brains and four eyes is a formula that works,” said Harter.

Critics also ask how AI will cope with unexpected events, like the substation fire that saw London's Heathrow Airport plunged into darkness on 20 March. A sudden loss of ground data – from runway lights to landing localiser signals – could affect safe operation by an AI hungry for data to crunch.

A prime example of such an unexpected aviation event was the Qantas flight QF32 incident off Singapore in November 2010, cited by many RAeS speakers as a talismanic warning for SPO. When an engine exploded on QF32, debris severed 400 network cables and holed a fuel tank. The Airbus A380's flight computer, starved of accurate sensor data, instructed the pilots to transfer fuel from an unholed wing tank into the holed one – risking serious fuel loss. The pilots ignored the demand, but if that computer had been flying the plane, calamity could have ensued.

The furore over SPO has seen the European Union Aviation Safety Agency (EASA), which was asked by plane-makers Airbus and Dassault Aviation to investigate SPO's potential in 2022, push back any regulatory orders, with the idea set to be revisited after 2027. "We have refocused our plans to increase the level of safety on flight decks with new concepts of operations," EASA official Pedro Caetano told the RAeS conference.

Ultimately, the fate of SPO will be down to you and me, air transport economist Floris de Haan told attendees. "The power of the consumer will decide if SPO goes ahead," he said. So, how should *New Scientist* readers make their views known, I asked him. "Airlines frequently do research on customer acceptance of new technology, so write to the airlines you are loyal to," he advises. You heard the man.